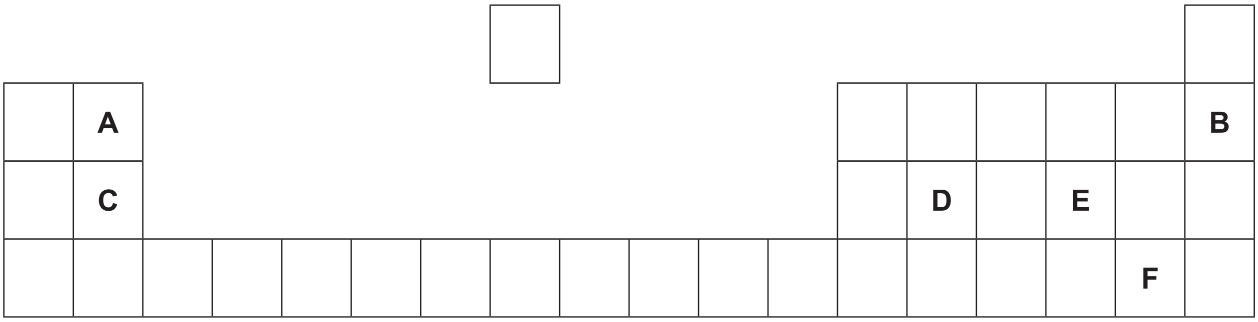
Answer **all** *questions*.

1. *(a)* The following diagram shows an outline of part of the Periodic Table.

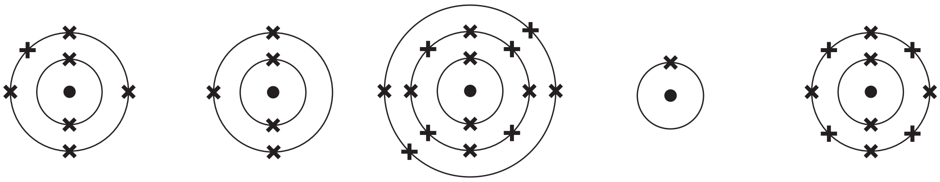
The letters shown are NOT the chemical symbols of the elements.



Choose **letters** from the diagram to complete the table below. [5]

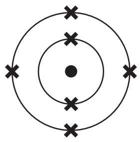
	Letter(s)
Two elements in the same group	.... and ....
The element which has 12 protons in its nucleus	....
The element with a full outer shell of electrons	....
The element in Group 2 and Period 2	....
The element with the electronic structure 2,8,4	....

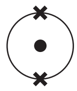
(b) Diagrams **1-5** show the electronic structure of five elements in the Periodic Table.



1 2 3 4 5

Give the **number** of the diagram which shows the electronic structure of the element which lies

(i) directly **below**  in the Periodic Table, ... [1]

(ii) to the **left** of  in the Periodic Table. .... [1]

(c) Nitrogen has two stable isotopes – nitrogen-14 and nitrogen-15.

Describe how these isotopes are similar to one another and how they are different. [2]

....

....

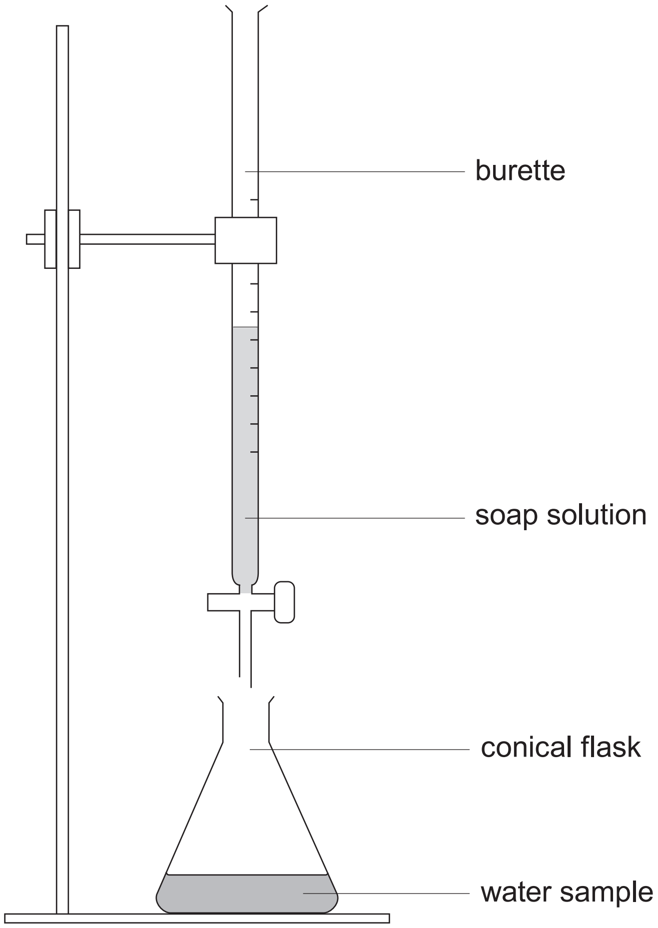
....

3410NA01 03

9

Answer **all** *questions.*

1. Water samples **A**, **B**, **C** and **D** were tested for hardness using the apparatus shown.



Soap solution was added 1 cm^3 at a time to each sample and the volume required to produce a permanent lather on shaking was recorded. Each sample was tested before and after boiling. The results are shown in the table.

Water sample	Volume of soap solution required (cm^3)	
	Before boiling	After boiling
A	1	1
B	10	10
C	15	1
D	15	8

Examiner only

(a) (i) State which water sample contains **only** temporary hardness. Explain your answer. [2]

Water sample ....

Explanation ....

....

....

(ii) Give **one** similarity in the composition of temporary and permanent hard water. [1]

....

....

(b) Discuss the benefits and drawbacks of living in a hard water area. [3]

....

....

....

....

....

....

3410UA01 03

6

Examiner only

Answer **all** *questions.*

1. *(a)* The following table shows some information about Group 1 elements.

Metal	Melting point (°C)	Boiling point (°C)	Density (g/cm^3)	Reaction with chlorine
lithium	180	1342	0.54	reacts slowly to make a white salt
sodium	97	883	0.97	burns vigorously with a yellow flame to make a white salt
potassium	63	759	0.88	reacts violently to make a white salt
rubidium	39	688	1.53	explosive reaction
caesium	28	671	1.93	explosive reaction

(i) Describe the trend in density going down the group. [1]

....

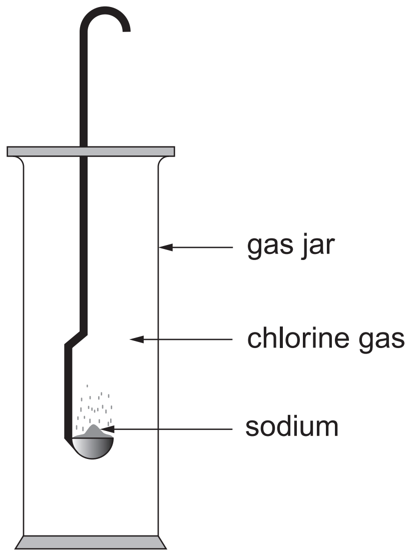
(ii) Explain the difference in reactivity down the group in terms of electronic structure. [2]

....

....

Examiner only

(b) The apparatus below can be used to demonstrate the reaction between sodium and chlorine.



(i) Apart from the use of safety goggles, state **one** safety precaution that needs to be followed when using **each** of these elements. [2]

Element	Safety precaution
sodium	....
chlorine	....

(ii) Complete and balance the symbol equation for the reaction that takes place between sodium and chlorine. [2]

$\square\ Na + Cl_2 \longrightarrow \square$ ....

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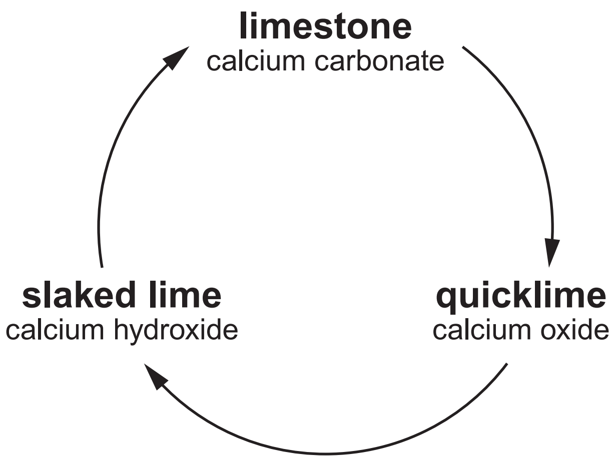
 (3410UA0-1) Turn over.

Examiner only

Answer **all** questions.

1. Limestone is a rock which consists mostly of calcium carbonate. The diagram shows a cycle of reactions involving limestone.



(a) (i) When limestone is heated, calcium carbonate is converted to calcium oxide and carbon dioxide.

I. State the name for this type of reaction. [1]

....

II. Write a balanced symbol equation for the reaction. [2]

.... $\longrightarrow$ .... + ...

(ii) State what must be added to calcium oxide to form calcium hydroxide. [1]

....

(b) Give **one** use of limestone in the construction industry. [1]

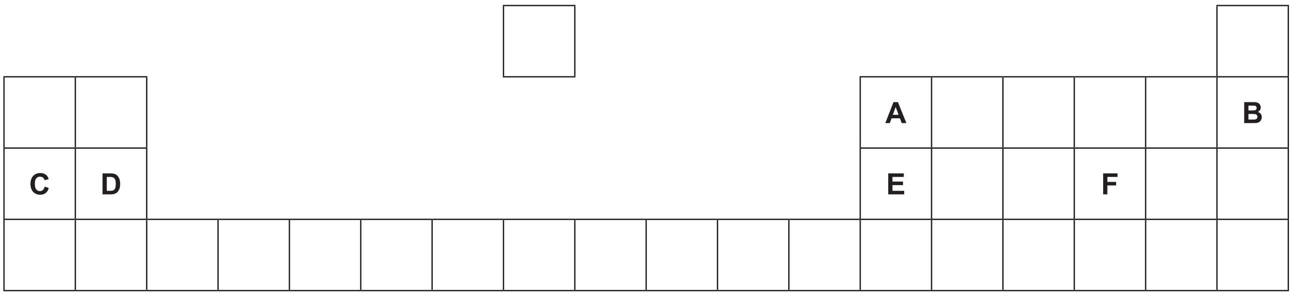
....

5

Answer **all** questions.

1. (a) The following diagram shows an outline of part of the Periodic Table.

The letters shown are NOT the chemical symbols of the elements.

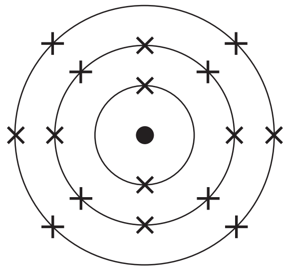


Choose **letters** from the diagram to complete the table below. [4]

	Letter
The element in Group 3 and Period 2	....
The element which has 10 protons in its nucleus	....
The element with the electronic structure 2,8,6	....
The element which forms a 2+ ion	....

Examiner only

(b) The diagram below shows the electronic structure of an element in the Periodic Table.



In the space below, draw a diagram to show the electronic structure of the element which lies directly **above** it. [1]

3410UA01 03

(c) The table shows information about atoms **X**, **Y** and **Z**.

Atom	Symbol	Number of protons	Number of neutrons	Number of electrons
X	$^{31}_{15}\mathbf{X}$	…………………………….	16	15
Y	$^{39}_{19}\mathbf{Y}$	19	…………………………….	19
Z	$^{40}_{19}\mathbf{Z}$	19	21	…………………………….

(i) Complete the table. [3]

(ii) Underline the term used to describe atoms **Y** and **Z**. [1]

ions **inert** **insoluble** **isotopes**

9

